

2009



Mathematics

Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using blue or black pen
- Write your Student Number on every page
- Begin each question in a new booklet.
- All necessary working must be shown.
- Marks may be deducted for careless or poorly presented work.
- Board-approved calculators may be used.
- A list of standard integrals is included at the end of this paper.
- The mark allocated for each question is listed at the side of the question.

Total Marks –

- Attempt ALL questions.
- All questions are of equal value.

Students are advised that this is a Trial Examination only and does not necessarily reflect the content or format of the Higher School Certificate Examination.

Question 1 – (15 marks) – Start a new booklet

Marks

a) Simplify i^{2009} 1

b) (i) Find real numbers x and y such that 2

$$x + iy = \sqrt{24 - 10i}$$

(ii) Solve the quadratic equation 2

$$z^2 + (1 - 3i)z - (8 - i) = 0$$

c) (i) Express $-\sqrt{3} + i$ in modulus-argument form. 2

(ii) Hence express $(-\sqrt{3} + i)^8$ in the form $a + bi$ where a and b are real numbers (in simplified form). 2

d) On an Argand diagram shade the region containing all points representing complex numbers, z , such that 3

$$2 \leq |z| \leq 3 \text{ and } \frac{-\pi}{3} < \arg z \leq \frac{2\pi}{3}$$

e) On separate diagrams draw a neat sketch of the locus specified by

(i) $\arg(z - 1 + i) = \frac{\pi}{4}$ 1

(ii) $\arg\left(\frac{z-1+i}{z-i}\right) = 0$ 2

Question 2 – (15 marks) – Start a new booklet

Marks

- a) Using the substitution $u = \sqrt{x^3 + 1}$ or otherwise find

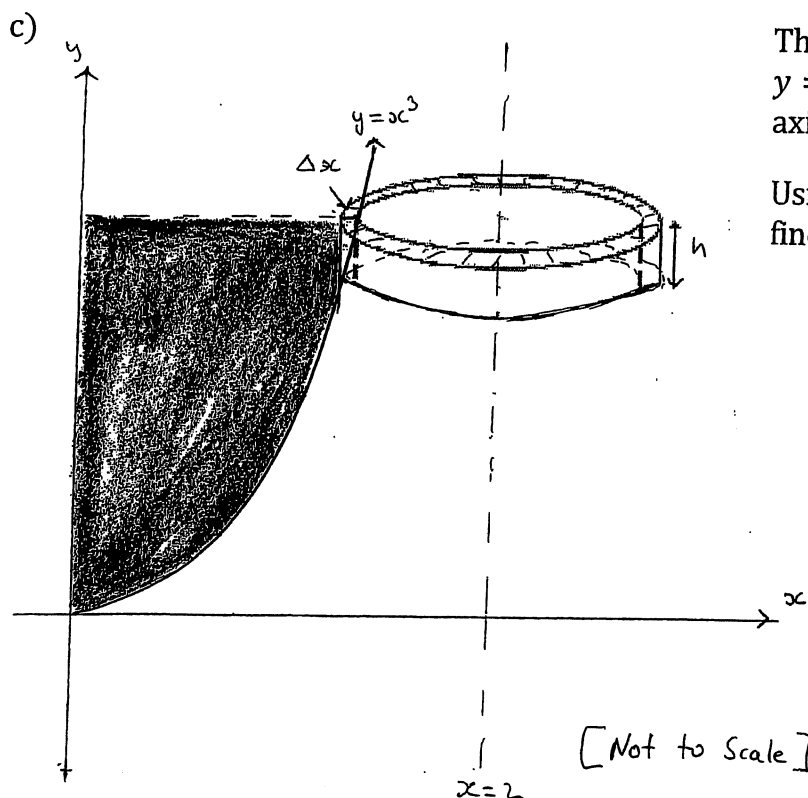
3

$$\int_0^2 \frac{x^5}{\sqrt{x^3 + 1}} dx$$

- b) By completing the square find

2

$$\int \frac{dx}{\sqrt{7 + 6x - x^2}}$$



3

- d) (i) Show that $\sin(A + B) + \sin(A - B) = 2 \sin A \cos B$

1

- (ii) Find all the solutions to the equation

3

$$\sin x + \sin 3x = \cos x$$

- e) Use the substitution $t = \tan \frac{\theta}{2}$ to find

3

$$\int_0^{\frac{\pi}{2}} \frac{d\theta}{1 + \cos \theta + 3 \sin \theta}$$

Question 3 – (15 marks) – Start a new booklet

Marks

a) The remainder when $x^4 + ax + b$ is divided by $(x + 3)(x - 2)$ is $x - 3$. Find the values of a and b . 2

b) $z = 1 - i$ is a root of the equation $z^3 + mz^2 + nz + 6 = 0$ where m and n are real. 3
Find the values of m and n .

c) (i) Find the general solution of the equation $\cos 3\theta = \frac{1}{2}$ 1

(ii) Show that $\cos 3\theta = 4\cos^3 \theta - 3\cos \theta$ 2

(iii) Using the substitution $x = \cos \theta$, and part (ii), express the equation in (i) as a polynomial in terms of x . 1

(iv) Hence, show that $\cos \frac{\pi}{9} + \cos \frac{5\pi}{9} + \cos \frac{7\pi}{9} = 0$ 2

(v) Find the polynomial of least degree that has zeros 2

$$\left(\sec \frac{\pi}{9}\right)^2, \left(\sec \frac{5\pi}{9}\right)^2, \left(\sec \frac{7\pi}{9}\right)^2$$

d) Find: 2

$$\int x \cdot e^{2x} dx$$

Question 4 – (15 marks) – Start a new booklet

Marks

- a) State whether the following is True or False. Give a brief reason.

1

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \tan^7 \theta \, d\theta > 0$$

[Note: You are not required to find the primitive function]

- b) The hyperbola H has equation

$$\frac{x^2}{9} - \frac{y^2}{16} = 1$$

- (i) Find the eccentricity of H and hence write down the coordinates of the foci, S and S' , and the equations of the directrices.

3

- (ii) Write down the equations of the asymptotes of H .

1

- (iii) Sketch H , clearly showing the foci, directrices and asymptotes.

2

- (iv) $P(3 \sec \theta, 4 \tan \theta)$ is a point on H . Prove that the tangent at P has equation

2

$$\frac{x \sec \theta}{3} - \frac{y \tan \theta}{4} = 1$$

- (v) This tangent cuts the asymptotes at A and B . Prove that

(α) $PA = PB$ and

3

- (β) the area of $\triangle OAB$ is independent of the position of P on the hyperbola.

3

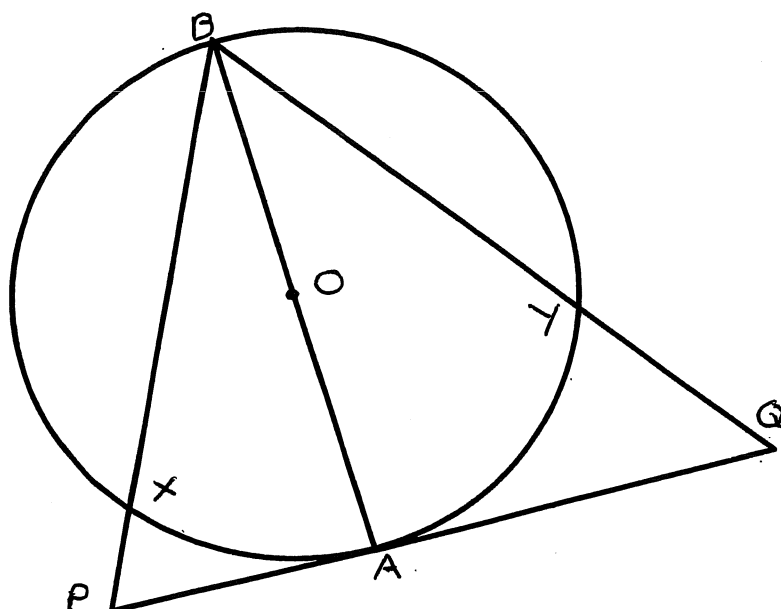
Question 5 – (15 marks) – Start a new booklet

Marks

- a) Find the equation of the tangent to the curve $x^3 - 2xy + y^2 = 4$ at the point $(-2, 2)$

2

b)



PAQ is a tangent to the circle with centre O and AB is a diameter.

3

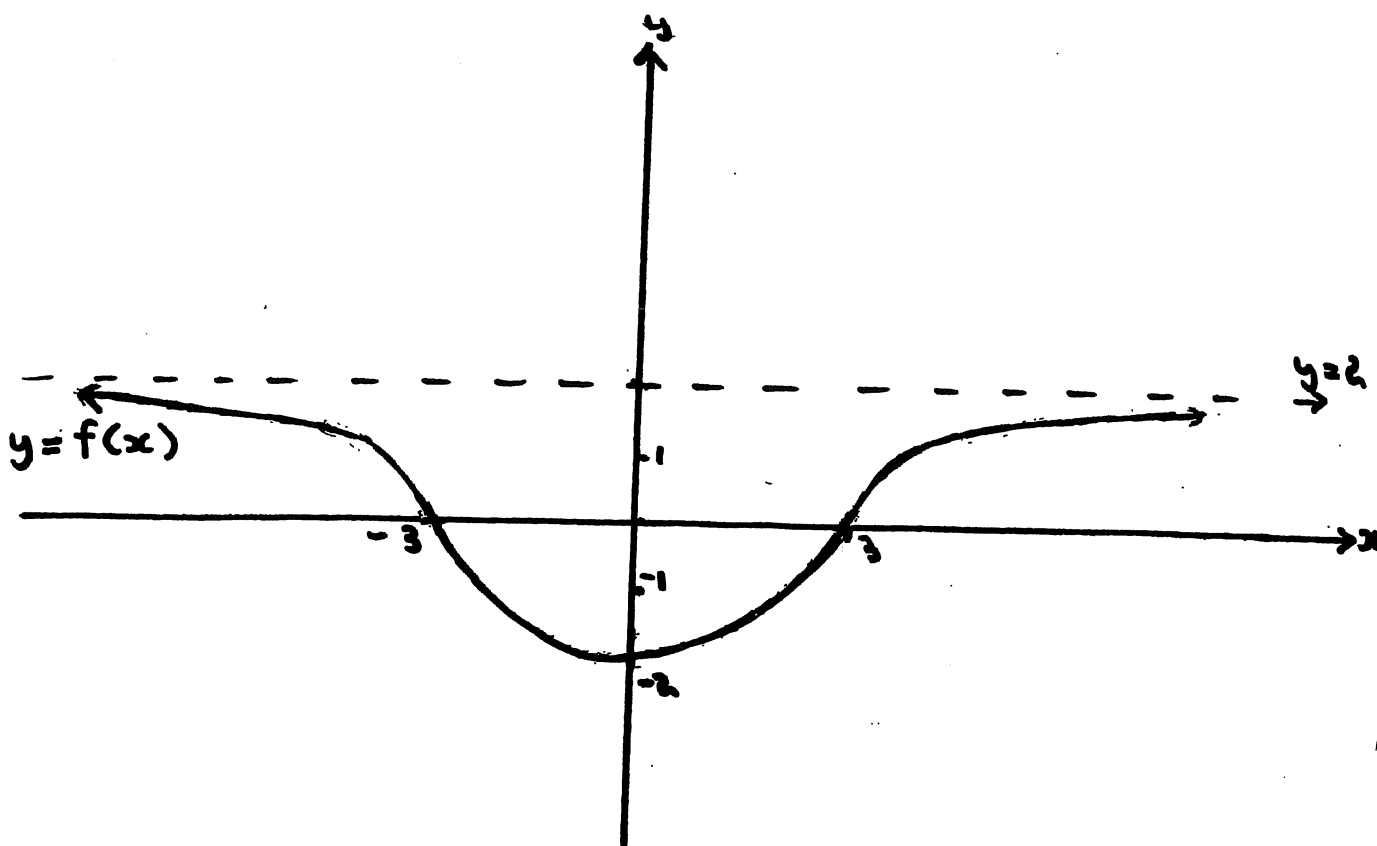
PB cuts the circle at X and QB cuts the circle at Y .

Prove that $PQYX$ is a cyclic quadrilateral.

Question 5 (cont'd)

Marks

c)



The graph of $y = f(x)$ is shown. On the answer sheets provided draw the graphs of the following:

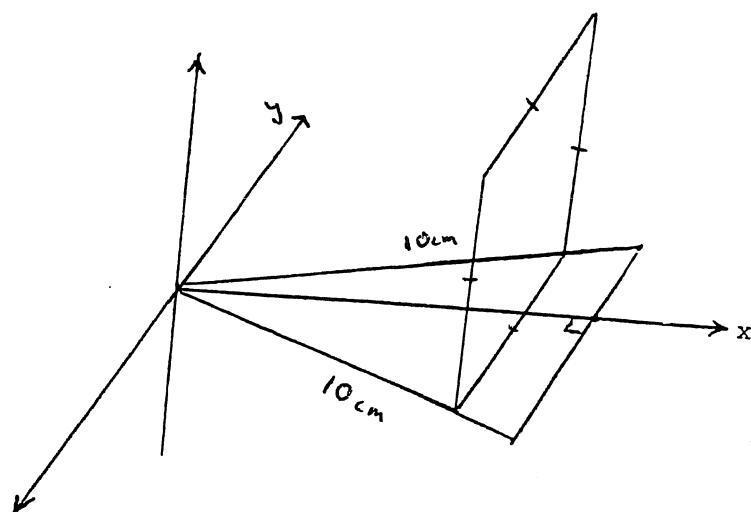
- | | |
|---------------------------|---|
| (i) $y = (f(x))^2$ | 2 |
| (ii) $y = f(x) $ | 2 |
| (iii) $y^2 = f(x)$ | 2 |
| (iv) $y = \frac{1}{f(x)}$ | 2 |
| (v) $y = f'(x)$ | 2 |

Question 6 – (15 marks) – Start a new booklet

Marks

- a) The base of a solid is an equilateral triangle of side length 10 cm, with one vertex at the origin and one side parallel to the y -axis as shown in the diagram.

Each cross-section perpendicular to the x -axis is a square with one side in the base of the solid.



- (i) Show that the area of the cross-section x cm from the origin is

2

$$A(x) = \frac{4x^2}{3}$$

- (ii) Hence, find the volume of the solid.

3

Question 6 (cont'd)

Marks

- b) A particle of mass m is projected vertically upwards in a medium where it experiences a resistance of magnitude mkv^2 where k is a positive constant and v is the velocity of the particle.

During the downward motion the terminal velocity of the particle is V . Its initial velocity of projection is $\frac{1}{5}$ of this terminal velocity.

- (i) By considering the forces on the particle during its downward motion, show that 2

$$kV^2 = g$$

(where g is the acceleration due to gravity)

- (ii) Show that during its upward motion the acceleration of the particle $\ddot{x}$ is given by 1

$$\ddot{x} = -g \left(1 + \frac{v^2}{V^2} \right)$$

- (iii) If the distance travelled by the particle in its upward motion is x when its velocity is v , show that the maximum height H reached is given by 3

$$H = \frac{V^2}{2g} \ln \left(\frac{26}{25} \right)$$

- (iv) If the velocity of the particle is v when it has fallen a distance of y from its maximum height, show that 2

$$y = \frac{V^2}{2g} \ln \left[\frac{V^2}{V^2 - v^2} \right]$$

- (v) The velocity of the particle is U when it returns to its point of projection. Show that 2

$$\frac{V}{U} = \sqrt{26}$$

Question 7 – (15 marks) – Start a new booklet

Marks

- a) (i) Prove that

2

$$\int_0^a f(a-x) dx = \int_0^a f(x) dx$$

- (ii) Hence evaluate

2

$$I = \int_0^1 \frac{x^{10}}{x^{10} + (1-x)^{10}} dx$$

- b) If $P(cp, \frac{c}{p})$ and $Q(cq, \frac{c}{q})$ are two points on the rectangular hyperbola $xy = c^2$

- (i) Show that the equation of the chord PQ is

2

$$x + pqy = c(p + q)$$

- (ii) If the chord passes through the point $R(a, b)$ prove that the locus of the mid point of the chord is given by

3

$$2xy = ay + bx$$

- c) (i) Use induction to prove that

3

$$1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

for positive integers $n \geq 1$

- (ii) Hence, or otherwise, find

3

$$2^2 + 5^2 + 8^2 + \dots + (3n-1)^2$$

Question 8 - (15 marks) – Start a new booklet

Marks

- a) ADB is a straight line with $AD = a$ and $DB = b$. A circle is drawn with AB as diameter. DC is drawn perpendicular to AB and meets the circle at C .

(i) By using similar triangles show that $DC = \sqrt{ab}$.

2

(ii) Deduce geometrically that if a and b are positive real numbers then

1

$$\sqrt{ab} \leq \frac{a+b}{2}$$

(iii) Using (ii), or otherwise, prove that if x, y, z are positive real numbers then

2

$$(x+y)(y+z)(z+x) \geq 8xyz$$

- b) For a certain series the n th term is given by

$$T_n = x^{n-1}(1 + x + x^2 + \dots + x^{n-1})$$

(i) Show that S_n , the sum to n terms, of this series is given by

3

$$S_n = \frac{(1-x^n)(1-x^{n+1})}{(1-x)(1-x^2)} \quad \text{provided } x^2 \neq 1$$

(ii) Deduce that

2

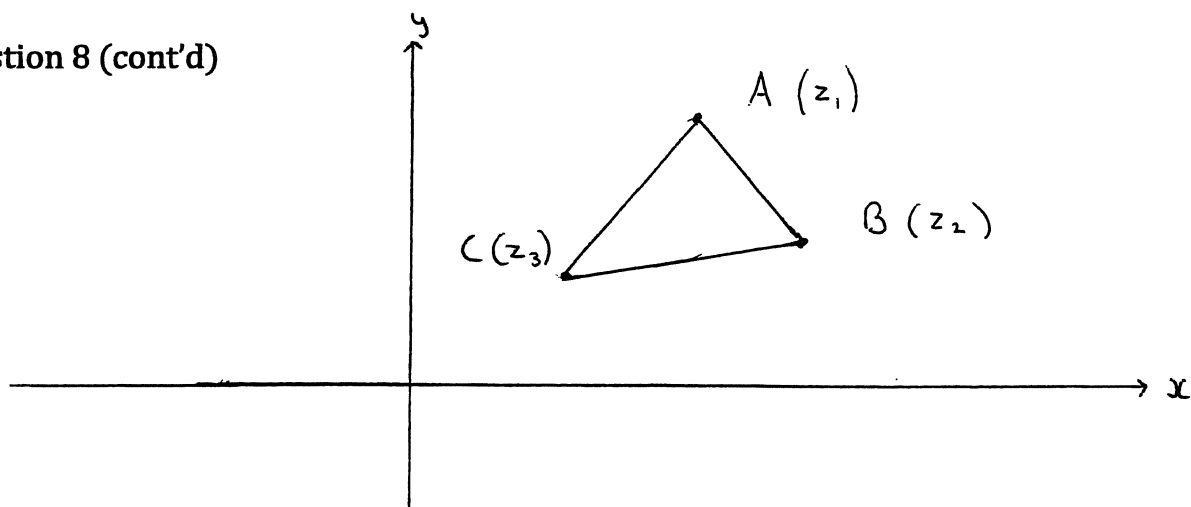
$$\lim_{x \rightarrow 1} \frac{(1-x^n)(1-x^{n+1})}{(1-x)(1-x^2)} = \frac{1}{2} n(n+1)$$

Question 8 (cont'd)

Marks

c)

5



A, B and C are the points that represent the complex numbers z_1, z_2, z_3 on the Argand diagram

Prove that if

$$\frac{z_2 - z_3}{z_1 - z_3} = \frac{z_1 - z_3}{z_1 - z_2}$$

then $\triangle ABC$ is equilateral.

Question 1

$$\begin{aligned} (a) \quad i^{2009} &= (i^4)^{502} \times i \\ &= 1^{502} \times i \\ &= i \end{aligned} \quad (1)$$

$$\begin{aligned} (b)(i) \quad (x+iy)^2 &= 24-10i \\ x^2-y^2 &= 24 \quad (1) \\ 2xyi &= -10i \\ xy &= -5 \\ y &= -\frac{5}{x} \quad (2) \end{aligned}$$

Subst (2) in (1)

$$x^2 - \frac{25}{x^2} = 24$$

$$x^4 - 24x^2 - 25 = 0 \quad - (1)$$

$$(x^2-25)(x^2+1) = 0$$

$$(x-5)(x+5)(x^2+1) = 0$$

$$x = 5, -5 \quad (x \in \mathbb{R})$$

$$y = -1, 1$$

$$\sqrt{24-10i} = \pm(5-i) \quad - (1)$$

$$(ii) \quad z^2 + (1-3i)z - (8-i) = 0$$

$$\begin{aligned} \Delta &= (1-3i)^2 - 4 \times 1 \times -(8-i) \\ &= 1-6i+9i^2+32-4i \\ &= 24-10i \end{aligned}$$

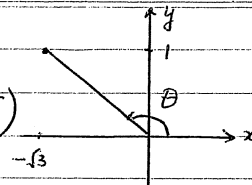
$$z = \frac{-(1-3i) \pm \sqrt{24-10i}}{2} \quad - (1)$$

$$= \frac{-1+3i \pm (5-i)}{2}$$

$$= \frac{4+2i}{2}, \frac{-6+4i}{2} = 2+i, -3+2i \quad - (1)$$

$$(c)(i) \quad -\sqrt{3} + i$$

$$= 2 \left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \right)$$



$$\begin{aligned} |-\sqrt{3}+i|^2 &= (\sqrt{3})^2 + 1^2 \\ &= 4 \end{aligned}$$

$$|-\sqrt{3}+i| = 2 \quad - (1)$$

$$\arg(-\sqrt{3}+i) = \theta$$

$$\tan \theta = -\frac{1}{\sqrt{3}}$$

$$\theta = \frac{5\pi}{6} \quad - (1)$$

$$\begin{aligned} (ii) \quad (-\sqrt{3}+i)^8 &= 2^8 \left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \right)^8 \\ &= 256 \left(\cos \frac{40\pi}{6} + i \sin \frac{40\pi}{6} \right) \end{aligned}$$

$$\frac{40\pi}{6} = \frac{20\pi}{3}$$

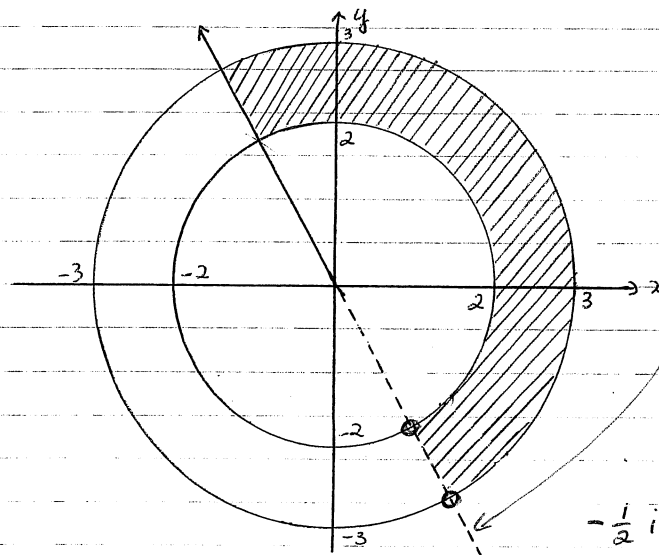
$$= 6\pi + \frac{2\pi}{3}$$

$$= 256 \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right) \quad - (1)$$

$$= 256 \left(-\frac{1}{2} + i \frac{\sqrt{3}}{2} \right)$$

$$= -128 + 128\sqrt{3}i \quad - (1)$$

(d)



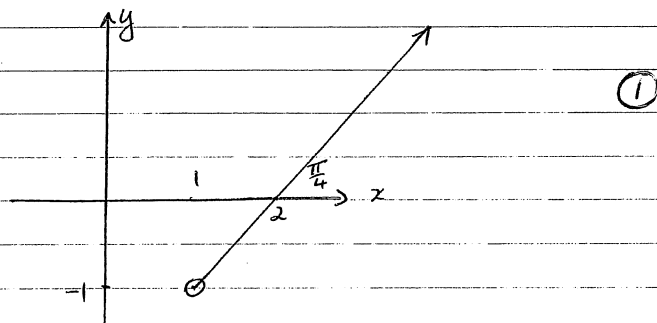
1 for line
(dotted & solid)
1 for annulus
1 for intersection

$-\frac{1}{2}$ if open circles missing

(3)

(e) (i) $\arg(3-1+i) = \frac{\pi}{4}$

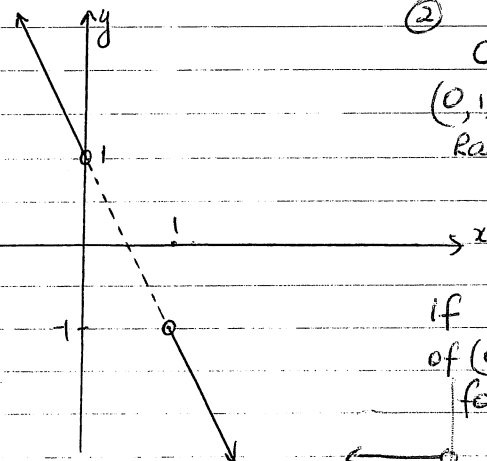
$\arg(3-(1-i)) = \frac{\pi}{4}$



(ii) $\arg\left(\frac{3-1+i}{3-i}\right) = 0$

$\arg(3-(1-i)) = \arg(3-i) = 0$

$\arg(3-(1-i)) = \arg(3-i)$



Open circles at
(0,1) and (1,-1)
Rays as shown

if used (0,-1) instead
of (0,1) could get ①
for



Question 2

(a) $\int_0^2 \frac{x^5}{\sqrt{x^3+1}} dx$

$u = (x^3+1)^{1/2}$
 $du = \frac{1}{2}(x^3+1)^{-1/2} \cdot 3x^2 dx$

$= \frac{3x^2}{2\sqrt{x^3+1}} dx$

When $x=0$ $u=1$

$x=2$ $u=3$

$\frac{1}{2} \int_0^2 \frac{x^3 \cdot 3x^2}{2\sqrt{x^3+1}} dx$

$= \frac{2}{3} \int_1^3 u^2 - 1 du$

$= \frac{2}{3} \left[\frac{u^3}{3} - u \right]_1^3$

$= \frac{2}{3} \left\{ \left(\frac{27}{3} - 3 \right) - \left(\frac{1}{3} - 1 \right) \right\}$

$= \frac{40}{9}$

OR $\int_0^2 \frac{x^5}{\sqrt{x^3+1}} dx$

$u = \sqrt{x^3+1}$
 $u^2 = x^3+1$

$2u du = 3x^2 dx$

$x=0$ $u=1$

$x=2$ $u=3$

$\frac{1}{3} \int_1^3 \frac{x^3 \cdot 3x^2}{\sqrt{x^3+1}} dx$

$= \frac{1}{3} \int_1^3 \frac{(u^2-1) \cdot 2u du}{u}$

$= \frac{2}{3} \int_1^3 u^2 - 1 du$ (then as above)

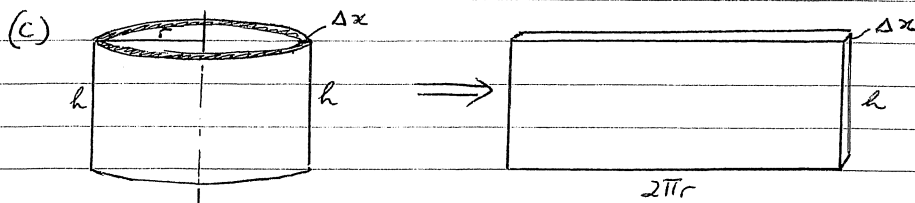
(b) $7+6x-x^2 = 7-(x^2-6x+9-9)$
 $= 16-(x-3)^2$

* Completing Square
poorly done.

①

$\int \frac{dx}{\sqrt{16-(x-3)^2}} = \sin^{-1} \frac{(x-3)}{4} + C$

①



Volume of cylindrical shell = ΔV

$$\Delta V \doteq 2\pi r h \Delta x$$

$$= 2\pi (2-x)(1-x^3) \Delta x$$

$$h = 1-y = 1-x^3$$

$$r = 2-x$$

* Most missed part was $(1-y)$ for height.

$$V = \lim_{\Delta x \rightarrow 0} \sum_{x=0}^1 \Delta V$$

$$= \lim_{\Delta x \rightarrow 0} \sum_{x=0}^1 2\pi (2-x-2x^3+x^4) \Delta x$$

$$= 2\pi \int_0^1 (2-x-2x^3+x^4) dx$$

$$= 2\pi \left[2x - \frac{x^2}{2} - \frac{2x^4}{4} + \frac{x^5}{5} \right]_0^1$$

$$= 2\pi \left\{ \left(2 - \frac{1}{2} - \frac{1}{2} + \frac{1}{5} \right) - 0 \right\}$$

$$= 2\pi \times \frac{6}{5}$$

$$\text{Volume} = \frac{12\pi}{5} \text{ units}^3$$

$$\begin{aligned} (d)(i) \sin(A+B) + \sin(A-B) &= \sin A \cos B + \cos A \sin B + \sin A \cos B - \cos A \sin B \\ &= 2 \sin A \cos B \end{aligned}$$

$$\begin{aligned} (ii) \sin x + \sin 3x &= 2 \sin 2x \cos x \end{aligned}$$

$$\text{Let } A+B = 3x$$

$$A-B = x$$

$$2A = 4x \quad A = 2x$$

$$2B = 2x \quad B = x$$

$$\textcircled{\frac{1}{2}} A=2x \quad B=x$$

$$\sin x + \sin 3x = \cos x$$

$$2 \sin 2x \cos x - \cos x = 0$$

$$\cos x (2 \sin 2x - 1) = 0$$

$$\cos x = 0 \quad \text{or} \quad \sin 2x = \frac{1}{2}$$

$$x = \frac{(2k+1)\pi}{2}$$

$$2x = \frac{\pi}{6} + 2k\pi \quad \text{or} \quad \frac{5\pi}{6} + 2k\pi$$

$$k \in \mathbb{Z}$$

$$x = \frac{\pi}{12} + k\pi \quad \text{or} \quad \frac{5\pi}{12} + k\pi$$

$$x = \frac{(2k+1)\pi}{2}, \frac{\pi}{12} + k\pi, \frac{5\pi}{12} + k\pi \quad k \in \mathbb{Z}$$

$$(e) \int_0^{\frac{\pi}{2}} \frac{d\theta}{1 + \cos \theta + 3 \sin \theta}$$

$$= \int_0^1 \frac{1}{1 + \frac{1-t^2}{1+t^2} + \frac{6t}{1+t^2}} \cdot \frac{2}{1+t^2} dt$$

$$= \int_0^1 \frac{2}{1+t^2+1-t^2+6t} dt$$

$$= \int_0^1 \frac{2}{2+6t} dt$$

$$= \int_0^1 \frac{1}{1+3t} dt$$

$$= \left[\frac{1}{3} \ln(1+3t) \right]_0^1$$

$$= \frac{1}{3} (\ln 4 - \ln 1)$$

$$= \frac{\ln 4}{3}$$

$$\left(= \frac{2 \ln 2}{3} \right)$$

$$t = \tan \frac{\theta}{2}$$

$$\theta = 2 \tan^{-1} t$$

$$d\theta = \frac{2}{1+t^2} dt$$

$$\text{When } \theta = 0 \quad t = 0$$

$$\theta = \frac{\pi}{2} \quad t = 1$$

$$1 + \cos \theta + 3 \sin \theta$$

$$= 1 + \frac{1-t^2}{1+t^2} + \frac{3 \cdot 2t}{1+t^2}$$

* $\textcircled{-1}$ if carried error made integral easier

Question 3

(a) $x^4 + ax + b = (x+3)(x-2)(x) + (x-3)$

Subst $x = -3$

$$81 - 3a + b = -6$$

$$3a - b = 87 \quad \text{①}$$

$$2a + b = -17 \quad \text{②}$$

① + ② $5a = 70$

$$a = 14$$

Subst in ① $42 - b = 87$

$$b = -45$$

Subst $x = 2$

$$16 + 2a + b = -1$$

$$2a + b = -17$$

($\frac{1}{2}$ off for errors minor)

→ ①

(b) $z^3 + mz^2 + nz + 6 = 0$ has $z = 1-i$ as a root

$$(1-i)^2 = 1 - 2i + i^2 = -2i$$

$$(1-i)^3 = (1-i) \times -2i = -2 - 2i$$

$$\therefore -2 - 2i + m(-2i) + n(1-i) + 6 = 0$$

$$-2 + n + 6 + i(-2 - 2m - n) = 0$$

Equating real and imaginary parts:

$$n + 4 = 0$$

$$n = -4$$

$$-2 - 2m - n = 0$$

$$-2 - 2m + 4 = 0$$

$$2m = 2$$

$$m = 1$$

→ ①

→ 1

→ 1

OR Since the coefficients are real $1+i$ is also a root

Let the 3rd root be β

$$(1-i)(1+i)\beta = -6$$

$$2\beta = -6$$

} → ①

$$-m = \text{sum of roots}$$

$$= 1-i + 1+i + -3$$

$$= -1$$

$$m = 1$$

→ ①

$$n = \text{sum in pairs}$$

$$= (1-i)(1+i) + -3(1-i) + -3(1+i)$$

$$= 2 - 3(1-i + 1+i)$$

$$= 2 - 6$$

$$= -4$$

→ ①

(c) (i) $\cos 3\theta = \frac{1}{2}$

$$3\theta = \frac{\pi}{3} + 2k\pi, -\frac{\pi}{3} + 2k\pi$$

→ ①

$$\theta = \frac{2k\pi}{3} \pm \frac{\pi}{9}$$

$$\frac{\pi}{9} (6k \pm 1)$$

(ii) $\cos 3\theta = \cos(2\theta + \theta)$

$$= \cos 2\theta \cos \theta - \sin 2\theta \sin \theta$$

$$= (2\cos^2 \theta - 1)\cos \theta - 2\sin \theta \cos \theta \sin \theta$$

$$= 2\cos^3 \theta - \cos \theta - 2(1-\cos^2 \theta)\cos \theta$$

$$= 2\cos^3 \theta - \cos \theta - 2\cos \theta + 2\cos^3 \theta$$

$$= 4\cos^3 \theta - 3\cos \theta$$

} → ①

} → ①

(iii) $4\cos^3 \theta - 3\cos \theta = \frac{1}{2}$

$$4x^3 - 3x = \frac{1}{2}$$

$$8x^3 - 6x - 1 = 0$$

$$x = \cos \theta$$

→ ①

(iv) Roots of this cubic equation are

$$x = \cos \theta \text{ where } \theta = \frac{\pi}{9}, \frac{7\pi}{9}, \frac{5\pi}{9}$$

Note: $\theta = \frac{2k\pi}{3} + \frac{\pi}{9}$ gives $\frac{\pi}{9}, \frac{7\pi}{9}, \frac{13\pi}{9}, \dots$ $k=0,1,2$

$\theta = \frac{2k\pi}{3} - \frac{\pi}{9}$ gives $-\frac{\pi}{9}, \frac{5\pi}{9}, \frac{11\pi}{9}, \dots$ $k=0,1,2$

$\cos(\frac{\pi}{9}) = \cos(-\frac{\pi}{9})$; $\cos \frac{7\pi}{9} = \cos \frac{11\pi}{9}$; $\cos \frac{13\pi}{9} = \cos \frac{5\pi}{9}$

$\therefore$ Roots are $\cos \frac{\pi}{9}, \cos \frac{5\pi}{9}, \cos \frac{7\pi}{9}$

Sum of roots = $\frac{-\text{coeff } x^2}{\text{coeff } x^3}$
= 0

$\therefore \cos \frac{\pi}{9} + \cos \frac{5\pi}{9} + \cos \frac{7\pi}{9} = 0$

(v) Let α, β, γ be the roots of $8x^3 - 6x - 1 = 0$
Require the polynomial with roots $\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}$

Let $P(x) = 8x^3 - 6x - 1$

Required equation is

$P(\frac{1}{\sqrt{x}}) = 0$

$8(\frac{1}{\sqrt{x}})^3 - 6 \cdot \frac{1}{\sqrt{x}} - 1 = 0$

$\frac{8}{x\sqrt{x}} - \frac{6}{\sqrt{x}} - 1 = 0$

$8 - 6x - x\sqrt{x} = 0$

$8 - 6x = x\sqrt{x}$

$64 - 96x + 36x^2 = x^3$

$x^3 - 36x^2 + 96x - 64 = 0$

(d) $\int x e^{2x} dx = \int x \cdot \frac{d}{dx}(e^{2x}) dx$

$= \frac{1}{2} x e^{2x} - \int \frac{1}{2} e^{2x} dx$

$= \frac{1}{2} x e^{2x} - \frac{1}{4} e^{2x} + C$

Question 4

(a) $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \tan^7 \theta d\theta > 0$ False $(\frac{1}{2})$

$f(\theta) = (\tan \theta)^7$ is an odd function

$f(-\theta) = (\tan(-\theta))^7$

$= (-\tan \theta)^7$

$= -(\tan \theta)^7$

$= -f(\theta)$

Hence $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \tan^7 \theta d\theta = 0$

(b) $\frac{x^2}{9} - \frac{y^2}{16} = 1$

(i) $b^2 = a^2(e^2 - 1)$

$16 = 9(e^2 - 1)$

$e^2 = \frac{16}{9} + 1$

$= \frac{25}{9}$

① $e = \frac{5}{3}$ ($e > 0$)

$ae = 3 \times \frac{5}{3} = 5$

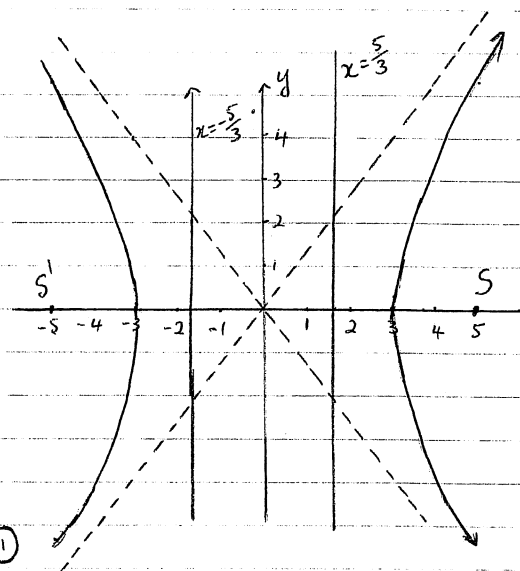
$\frac{a}{e} = \frac{3}{\frac{5}{3}} = \frac{9}{5}$

$S(5,0) S'(-5,0)$ ①

Directrices: $x = \pm \frac{5}{3}$ ①

(ii) $y = \pm \frac{4}{3} x$ ①

* Write equation of directrix & asymptote.



* Very poorly done.
Learn basics - check difference between Ellipse & Hyperbola going $0 < 0 > <$

(iv) $P(3\sec\theta, 4\tan\theta)$

$$\frac{x^2}{9} - \frac{y^2}{16} = 1$$

$$\frac{2x}{9} - \frac{2y}{16} \frac{dy}{dx} = 0$$

$$\frac{2x}{9} = \frac{2y}{16} \frac{dy}{dx}$$

$$\frac{2x}{9} \times \frac{16}{2y} = \frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{16x}{9y} \quad \left(\frac{1}{2}\right)$$

At P $\frac{dy}{dx} = \frac{16 \cdot 3\sec\theta}{9 \cdot 4\tan\theta}$
 $= \frac{4\sec\theta}{3\tan\theta} \quad \left(\frac{1}{2}\right)$

Eqⁿ of tangent is

$$y - 4\tan\theta = \frac{4\sec\theta}{3\tan\theta} (x - 3\sec\theta) \quad \left(\frac{1}{2}\right) \quad \left(\times \frac{\tan\theta}{4}\right)$$

$$\frac{y\tan\theta}{4} - \tan^2\theta = \frac{x\sec\theta}{3} - \sec^2\theta$$

$$\sec^2\theta - \tan^2\theta = \frac{x\sec\theta}{3} - \frac{y\tan\theta}{4} \quad \left(\frac{1}{2}\right)$$

$$\frac{x\sec\theta}{3} - \frac{y\tan\theta}{4} = 1$$

(v) When $y = \frac{4}{3}x$: $\frac{x\sec\theta}{3} - \frac{4x}{3} \cdot \frac{\tan\theta}{4} = 1$
 $\frac{x}{3} (\sec\theta - \tan\theta) = 1$

$$x = \frac{3}{\sec\theta - \tan\theta} \quad \left(\frac{1}{2}\right)$$

$$y = \frac{4}{3} \cdot \frac{3}{\sec\theta - \tan\theta}$$

$$= \frac{4}{\sec\theta - \tan\theta} \quad \left(\frac{1}{2}\right)$$

* Algebra poor.

A has coords $\left(\frac{3}{\sec\theta - \tan\theta}, \frac{4}{\sec\theta - \tan\theta}\right)$

When $y = -\frac{4}{3}x$ $\frac{x\sec\theta}{3} + \frac{4x}{3} \cdot \frac{\tan\theta}{4} = 1$

$$\frac{x}{3} (\sec\theta + \tan\theta) = 1$$

$$x = \frac{3}{\sec\theta + \tan\theta} \quad \left(\frac{1}{2}\right)$$

$$y = \frac{-4}{\sec\theta + \tan\theta} \quad \left(\frac{1}{2}\right)$$

B has coords $\left(\frac{3}{\sec\theta + \tan\theta}, \frac{-4}{\sec\theta + \tan\theta}\right)$

(2) Midpt of AB is :

$$x = \frac{1}{2} \left(\frac{3}{\sec\theta + \tan\theta} + \frac{3}{\sec\theta - \tan\theta} \right)$$

$$= \frac{3(\sec\theta - \tan\theta) + 3(\sec\theta + \tan\theta)}{2(\sec^2\theta - \tan^2\theta)}$$

$$= \frac{6\sec\theta}{2}$$

$$= 3\sec\theta \quad \left(\frac{1}{2}\right)$$

$$y = \frac{1}{2} \left(\frac{-4}{\sec\theta + \tan\theta} + \frac{4}{\sec\theta - \tan\theta} \right)$$

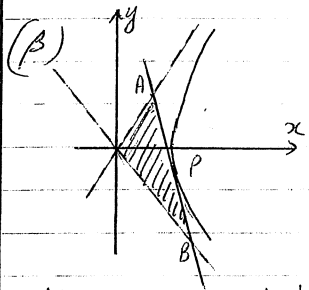
$$= \frac{1}{2} \left(\frac{-4\sec\theta + 4\tan\theta + 4\sec\theta + 4\tan\theta}{\sec^2\theta - \tan^2\theta} \right)$$

$$= \frac{8\tan\theta}{2 \times 1}$$

$$= 4\tan\theta \quad \left(\frac{1}{2} \right) \quad [3]$$

$$\therefore \text{Midpt of AB} = (3\sec\theta, 4\tan\theta) = P$$

ie P is the midpoint of AB ie $AP = BP$



$$\text{Area } \triangle AOB = \frac{1}{2} \times OA \times OB \times \sin \hat{AOB}$$

$$\hat{AOB} = 2 \times \hat{AOP}$$

$$= 2\theta \quad \left(\frac{1}{2} \right) \text{ where } \tan\theta = \frac{4}{3}$$

$$\sin \hat{AOB} = 2 \sin\theta \cos\theta \quad \sin\theta = \frac{4}{5}$$

$$= 2 \times \frac{4}{5} \times \frac{3}{5} \quad \cos\theta = \frac{3}{5}$$

$$= \frac{24}{25} \quad \left(\frac{1}{2} \right)$$

* Also done by $A = \frac{1}{2}bh$ using perpendicular distance formula.

* Not necessary to calculate $2 \tan\left(\frac{4}{3}\right) = \theta$

$$OA^2 = \frac{9}{(\sec\theta - \tan\theta)^2} + \frac{16}{(\sec\theta - \tan\theta)^2}$$

$$= \frac{25}{(\sec\theta - \tan\theta)^2}$$

$$OA = \frac{5}{|\sec\theta - \tan\theta|} \quad \left(\frac{1}{2} \right)$$

$$OB^2 = \frac{9 + 16}{(\sec\theta + \tan\theta)^2}$$

$$OB = \frac{5}{|\sec\theta + \tan\theta|} \quad [3]$$

$$\text{Area } \triangle AOB = \frac{1}{2} \times \frac{5}{|\sec\theta - \tan\theta|} \times \frac{5}{|\sec\theta + \tan\theta|} \times \frac{24}{25} \quad \left(\frac{1}{2} \right)$$

$$= \frac{12}{|\sec^2\theta - \tan^2\theta|} = \frac{12}{1} = 12 \quad \left(\frac{1}{2} \right)$$

Question 5

$$(a) \quad x^3 - 2xy + y^2 = 4$$

$$3x^2 - (2y + 2x \frac{dy}{dx}) + 2y \frac{dy}{dx} = 0$$

$$3x^2 - 2y = (2x - 2y) \frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{3x^2 - 2y}{2x - 2y} \quad \text{--- (1)}$$

$$\text{At } (-2, 2) \quad \frac{dy}{dx} = \frac{3(-2)^2 - 2 \times 2}{2(-2) - 2 \times 2}$$

$$= \frac{8}{-8}$$

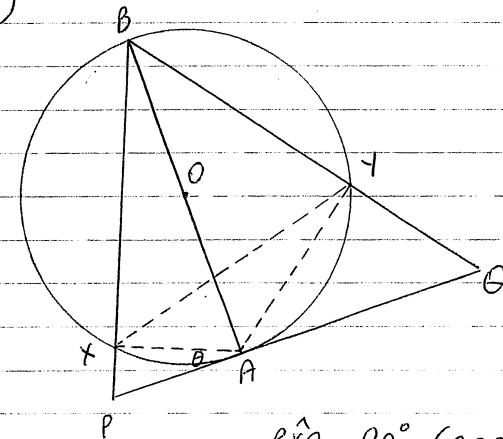
$$= -1$$

Equation of tangent is

$$y - 2 = -1(x + 2)$$

$$y = -x \quad \text{--- (1)}$$

(4)



Join AX, AY, XY

Let $\hat{PAX} = \theta$

$\therefore \hat{ABX} = \theta$

(angle between chord & tangent = angle in alternate segment)

$\hat{AYX} = \hat{ABX}$ (angles in same segment) = θ --- (1)

$\hat{BXA} = 90^\circ$ (angle in a semicircle)

$\therefore \hat{PXA} = 90^\circ$ ($\hat{BXP}$ is a straight angle)

$\therefore \hat{XPA} = 90^\circ - \theta$ (angle sum of $\triangle = 180^\circ$) --- (1)

Similarly $\hat{BYA} = \hat{OYA} = 90^\circ$

Hence $\hat{QYX} = 90^\circ + \theta$

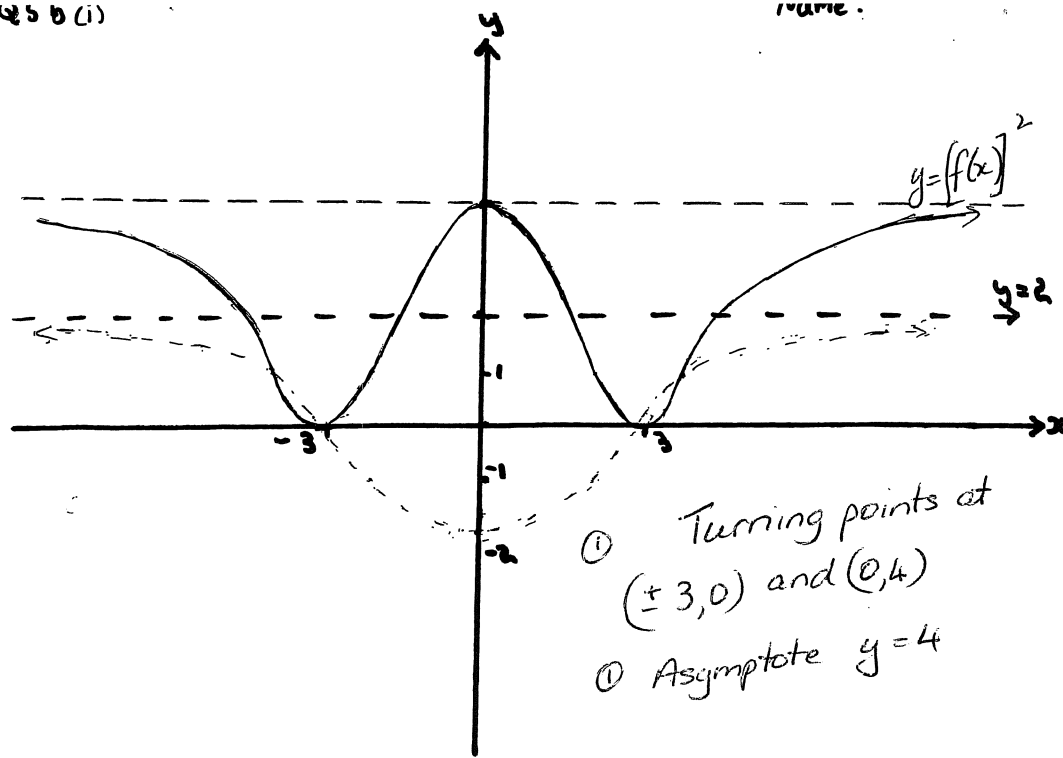
$$\therefore \hat{OPX} + \hat{QYX} = 90^\circ - \theta + 90^\circ + \theta \\ = 180^\circ$$

— ①

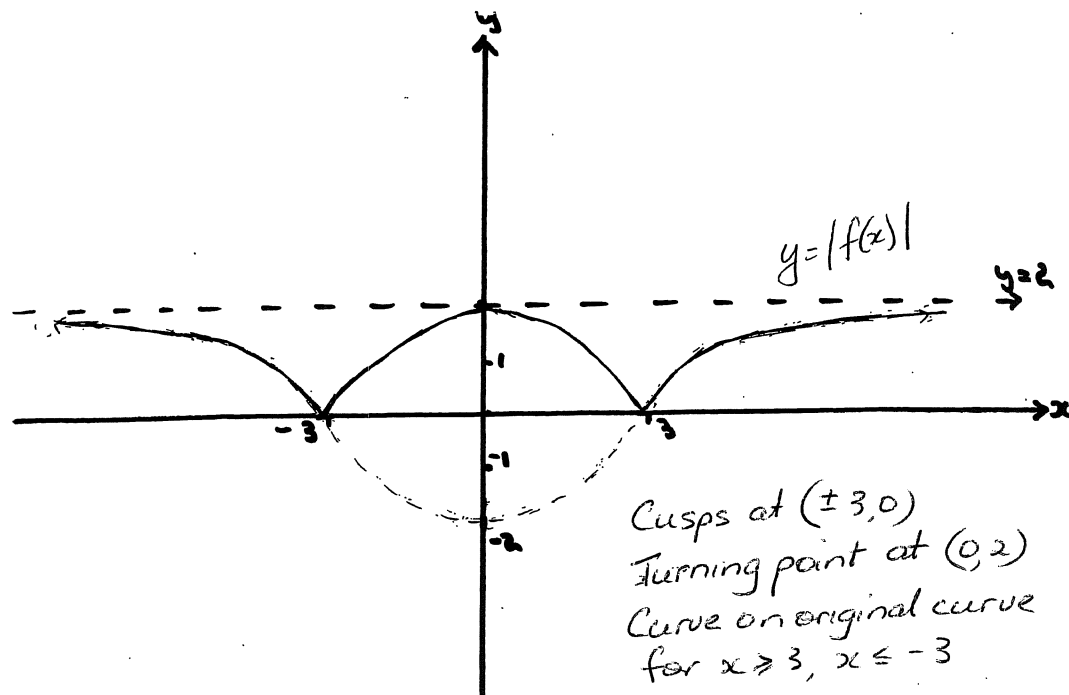
$\therefore POYX$ is a cyclic quadrilateral since opposite angles are supplementary

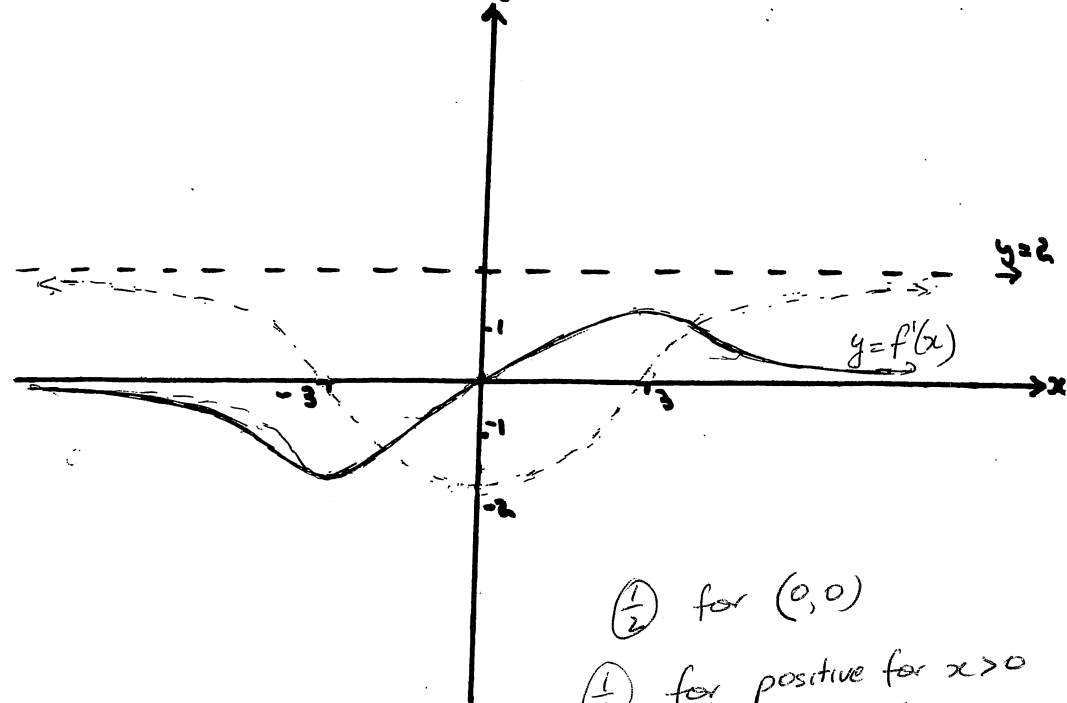
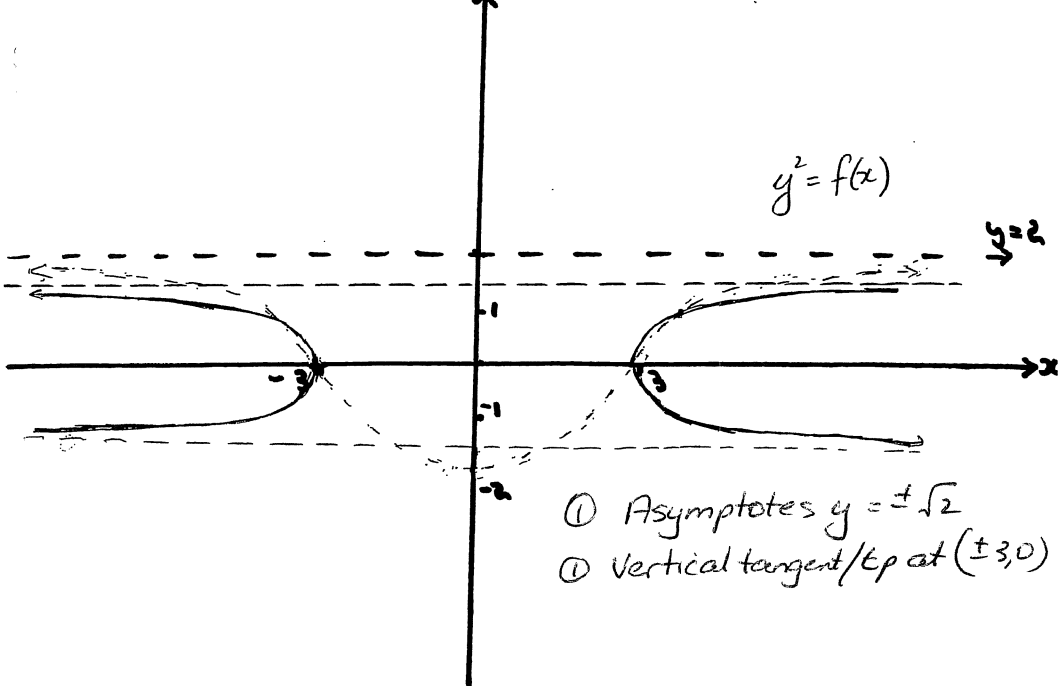
There are many methods to get to the result
— each was marked according to the correct logic displayed

Q5b(i)



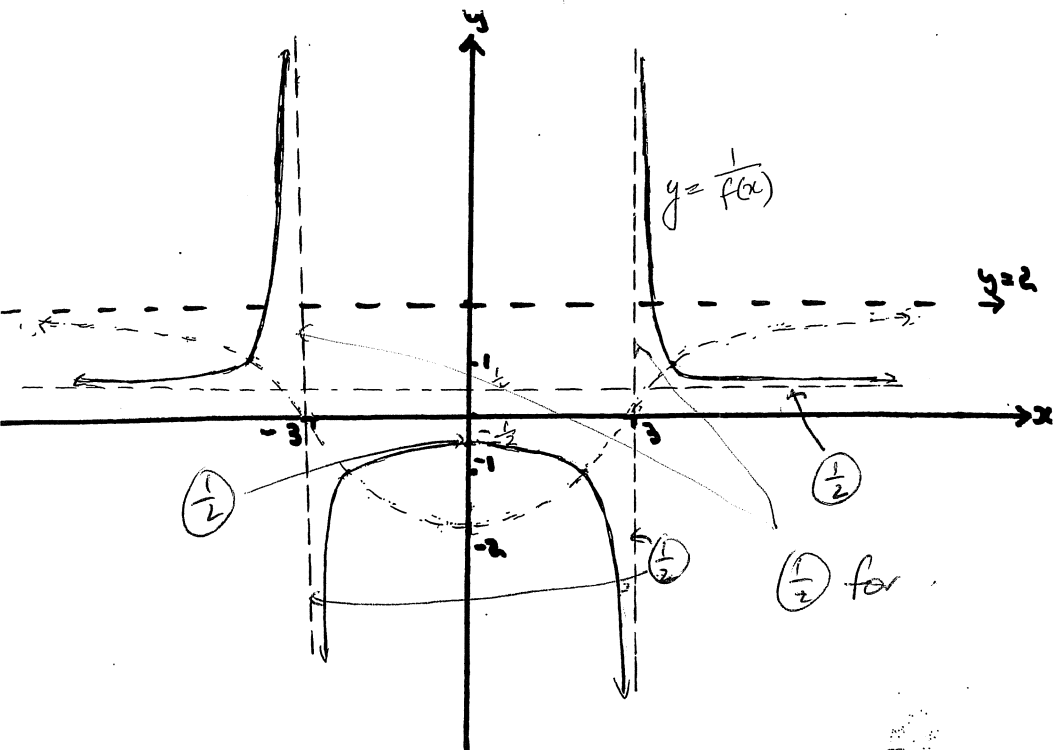
Q5b(ii)



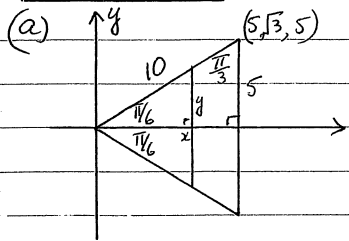


- ① for $(0,0)$
 ① for positive for $x > 0$
 negative for $x < 0$
 ① for $\rightarrow 0$ as $x \rightarrow \pm\infty$

5b (iv)



Question 6



When $y = 5$ $\frac{x}{5} = \tan \frac{\pi}{3}$
 $x = 5\sqrt{3}$ $\rightarrow$ ④

x ~~cm~~ from origin

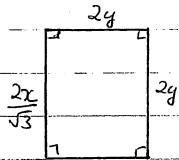
$$\frac{y}{x} = \tan \frac{\pi}{6}$$

$$y = \frac{x}{\sqrt{3}}$$

$$2y = \frac{2x}{\sqrt{3}}$$

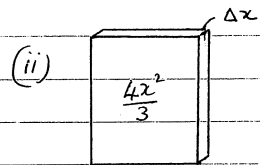
$$A(x) = (2y)^2$$

$$= \frac{4x^2}{3}$$



$$\rightarrow$$
 ①

$$\rightarrow$$
 ①



$$\Delta V = \frac{4x^2}{3} \Delta x$$

$$V = \lim_{\Delta x \rightarrow 0} \sum_{x=0}^{5\sqrt{3}} \frac{4x^2}{3} \Delta x$$

$$= \int_0^{5\sqrt{3}} \frac{4x^2}{3} dx$$

$$= \frac{4}{3} \left[\frac{x^3}{3} \right]_0^{5\sqrt{3}}$$

$$= \frac{4}{9} ((5\sqrt{3})^3 - 0)$$

$$= \frac{4 \times 125 \times 3\sqrt{3}}{9}$$

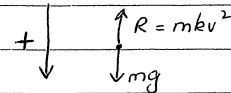
$$\text{Volume} = \frac{500\sqrt{3}}{3} \text{ cm}^3$$

$$\rightarrow$$
 ①

$$\rightarrow$$
 ①

$$\rightarrow$$
 ①

(u) (i) Downwards motion



$$m\ddot{x} = mg - mkv^2$$

$$\ddot{x} = g - kv^2$$

$$\rightarrow$$
 ①

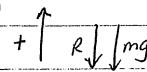
For terminal velocity $\ddot{x} \rightarrow 0$

$$0 = g - kv^2$$

$$kv^2 = g$$

$$\rightarrow$$
 ①

(ii) Upwards motion



$$m\ddot{x} = -mg - mkv^2$$

$$\ddot{x} = -(g + kv^2)$$

$$\rightarrow$$
 ②

$$= -g \left(1 + \frac{k}{g} v^2 \right)$$

$$= -g \left(1 + \frac{1}{v^2} v^2 \right)$$

$$= -g \left(1 + \frac{v^2}{v^2} \right)$$

$$\left. \begin{array}{l} \ddot{x} = -(g + kv^2) \\ \ddot{x} = -g \left(1 + \frac{k}{g} v^2 \right) \\ \ddot{x} = -g \left(1 + \frac{1}{v^2} v^2 \right) \\ \ddot{x} = -g \left(1 + \frac{v^2}{v^2} \right) \end{array} \right\} \rightarrow$$

(iii)

$$v \frac{dv}{dx} = -g \left(1 + \frac{v^2}{v^2} \right)$$

$$\frac{dv}{dx} = -g \left(\frac{v^2 + v^2}{v^2} \right)$$

$$\frac{dx}{dv} = -\frac{v^2 v}{g(v^2 + v^2)}$$

$$\rightarrow$$
 ①

$$x = -\frac{v^2}{2g} \ln(v^2 + v^2) + C$$

When $x = 0$ $v = \frac{V}{5}$

$$0 = -\frac{V^2}{2g} \ln \left(V^2 + \frac{V^2}{25} \right) + C$$

$$C = \frac{V^2}{2g} \ln \left(\frac{26V^2}{25} \right)$$

$$\rightarrow$$
 ①

$$x = \frac{v^2}{2g} \ln\left(\frac{26v^2}{25}\right) - \frac{v^2}{2g} \ln(v^2 + u^2)$$

When $u = 0$ $x = H$ (max height reached)

$$\begin{aligned} H &= \frac{v^2}{2g} \ln\left(\frac{26v^2}{25}\right) - \frac{v^2}{2g} \ln v^2 \\ &= \frac{v^2}{2g} \ln\left(\frac{26v^2}{25} \div v^2\right) \\ &= \frac{v^2}{2g} \ln\left(\frac{26}{25}\right) \end{aligned} \quad \rightarrow \textcircled{1}$$

($\frac{1}{2}$ off minor errors)

OR $H = \int_{\frac{v^2}{25}}^0 \frac{-v^2 u}{g(v^2 + u^2)} du$

$$\begin{aligned} &= \left[-\frac{v^2}{2g} \ln(v^2 + u^2) \right]_{\frac{v^2}{25}}^0 \\ &= -\frac{v^2}{2g} \ln v^2 + \frac{v^2}{2g} \ln\left(v^2 + \frac{v^2}{25}\right) \\ &= \frac{v^2}{2g} \left(\ln \frac{26v^2}{25} - \ln v^2 \right) \\ &= \frac{v^2}{2g} \ln\left(\frac{26}{25}\right) \end{aligned} \quad \rightarrow \textcircled{2}$$

(iv) Downwards motion

+
↓

$$\ddot{x} = g - kv^2$$

$$v \frac{dv}{dx} = g - kv^2$$

$$\frac{dv}{dx} = \frac{g - kv^2}{v}$$

$$\frac{dx}{dv} = \frac{v}{g - kv^2}$$

$$\frac{dx}{dv} = \frac{-1}{2k} \frac{-2kv}{g - kv^2}$$

$$x = -\frac{1}{2k} \ln(g - kv^2) + c \quad \rightarrow \textcircled{4}$$

When $x = 0$ $u = 0$

$$0 = -\frac{1}{2k} \ln g + c$$

$$c = \frac{1}{2k} \ln g$$

$$x = \frac{1}{2k} \ln g - \frac{1}{2k} \ln(g - kv^2)$$

$$= \frac{1}{2k} \ln\left(\frac{g}{g - kv^2}\right) \quad \rightarrow \textcircled{1}$$

$$= \frac{1}{2k} \ln\left(1 - \frac{1}{g} kv^2\right)$$

When $x = y$ velocity is v ; $\frac{1}{k} = \frac{v^2}{g}$
 $\frac{k}{g} = \frac{1}{v^2}$

$$\begin{aligned} y &= \frac{v^2}{2g} \ln\left(1 - \frac{1}{\frac{v^2}{g}} kv^2\right) \\ &= \frac{v^2}{2g} \ln\left(\frac{v^2}{v^2 - u^2}\right) \end{aligned} \quad \rightarrow \textcircled{1}$$

(v) When $y = H = \frac{v^2}{2g} \ln\left(\frac{26}{25}\right)$ $\left. \begin{array}{l} v = U \\ \end{array} \right\} \rightarrow \textcircled{1}$

$$\frac{v^2}{2g} \ln \frac{26}{25} = \frac{v^2}{2g} \ln\left(\frac{v^2}{v^2 - u^2}\right)$$

$$\frac{v^2}{v^2 - u^2} = \frac{26}{25}$$

$$\frac{1}{1 - \left(\frac{u}{v}\right)^2} = \frac{26}{25}$$

$$25 = 26 - 26\left(\frac{u}{v}\right)^2$$

$$\frac{26\left(\frac{u}{v}\right)^2}{26} = \frac{1}{26}$$

$$\left(\frac{v}{u}\right)^2 = 26$$

$$\frac{v}{u} = \sqrt{26}$$

→ ①

Question 7

$$(a)(i) \int_0^a f(a-x) dx$$

Let $u = a-x$ $\left(\frac{1}{2}\right)$
 $du = -dx$

$$= \int_a^0 f(u) \cdot (-1) du$$

When $x=0$ $u=a$

$x=a$ $u=0$ $\left(\frac{1}{2}\right)$

$$= \int_0^a f(u) du \quad \text{①}$$

$$= \int_0^a f(x) dx \quad \text{②}$$

$$(ii) \quad I = \int_0^1 \frac{x^{10}}{x^{10} + (1-x)^{10}} dx$$

$$= \int_0^1 \frac{(1-x)^{10}}{(1-x)^{10} + x^{10}} dx \quad (\text{from (i)})$$

$1 - (1-x) = x$

$$\therefore 2I = \int_0^1 \frac{x^{10}}{x^{10} + (1-x)^{10}} dx + \int_0^1 \frac{(1-x)^{10}}{(1-x)^{10} + x^{10}} dx$$

$$= \int_0^1 \frac{x^{10} + (1-x)^{10}}{x^{10} + (1-x)^{10}} dx$$

$$= \int_0^1 1 \cdot dx$$

$$= [x]_0^1$$

$$= 1 - 0$$

$$\therefore I = \frac{1}{2}$$

*Wrong number for a ① only

$$(4) (i) P(cp, \frac{c}{p}) \quad Q(cq, \frac{c}{q})$$

$$\begin{aligned} \text{Grad PQ} &= \frac{\frac{c}{p} - \frac{c}{q}}{cp - cq} \\ &= \frac{c \cdot \frac{q-p}{pq}}{c(p-q)} \\ &= -\frac{1}{pq} \quad (1) \end{aligned}$$

$\therefore$ Eqⁿ of PQ is

$$y - \frac{c}{p} = -\frac{1}{pq}(x - cp)$$

$$pqy - cq = -x + cp \quad (2)$$

$$x + pqy = c(p+q)$$

(ii) R(a, b) lies on PQ

$$a + pqb = c(p+q) \quad (3)$$

Let Midpt of PQ be (x, y)

$$\begin{aligned} x &= \frac{cp + cq}{2} & y &= \frac{1}{2} \left(\frac{c}{p} + \frac{c}{q} \right) \\ &= \frac{c(p+q)}{2} & &= \frac{c(p+q)}{2pq} \quad (4) \end{aligned}$$

$$2x = c(p+q)$$

$$\begin{aligned} \text{From (1)} \quad pq &= \frac{c(p+q) - a}{b} \\ &= \frac{2x - a}{b} \quad (5) \end{aligned}$$

$$\therefore \left(\frac{1}{2}\right)y = \frac{2x}{2[2x-a]}$$

$$2xy - ay = bx$$

$$2xy = ay + bx \quad (6)$$

$$(c) \text{ Aim to show } 1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\text{When } n=1 \quad \text{LHS} = 1^2 = 1$$

$$\text{RHS} = \frac{1(1+1)(2 \times 1 + 1)}{6} = 1 = \text{LHS} \quad \left(-\frac{1}{2}\right) \text{ not showing}$$

$\therefore$ Proposition is true for $n=1$ (1)

Let k be a positive integer for which proposition is true

$$\text{i.e. } 1^2 + 2^2 + 3^2 + \dots + k^2 = \frac{k(k+1)(2k+1)}{6}$$

Aim to show proposition is then true for $n=k+1$

$$\text{i.e. } 1^2 + 2^2 + 3^2 + \dots + k^2 + (k+1)^2 = \frac{(k+1)(k+1+1)(2(k+1)+1)}{6}$$

$$\begin{aligned} \text{LHS} &= 1^2 + 2^2 + \dots + k^2 + (k+1)^2 \\ &= \frac{k(k+1)(2k+1)}{6} + (k+1)^2 \quad \left(\frac{1}{2}\right) \end{aligned}$$

$$= \frac{(k+1)}{6} [k(2k+1) + 6(k+1)] \quad \left(\frac{1}{2}\right)$$

$$= \frac{(k+1)}{6} (2k^2 + 7k + 6)$$

$$= \frac{(k+1)(k+2)(2k+3)}{6} \quad \left(\frac{1}{2}\right)$$

$$= \text{RHS}$$

$\therefore$ Proposition is true for $n=k+1$ if true for $n=k$ etc

$\left(\frac{1}{2}\right)$ for here or at end showing $(k+1)(k+1+1)(2k+1+1)$

3

$$(ii) 2^2 + 5^2 + 8^2 + \dots + (3n-1)^2$$

$$= \sum_{k=1}^n (3k-1)^2 \quad \left(\frac{1}{2}\right)$$

$$= \sum_{k=1}^n (9k^2 - 6k + 1)$$

$$= 9 \sum_{k=1}^n k^2 - 6 \sum_{k=1}^n k + \sum_{k=1}^n 1 \quad \left(\frac{1}{2}\right)$$

$$= 9 \frac{n(n+1)(2n+1)}{6} - 6 \frac{n(n+1)}{2} + n \quad \left(\frac{1}{2}\right)$$

$$= \frac{3n(n+1)(2n+1) - 6n(n+1) + 2n}{2}$$

$$= \frac{n [3(n+1)(2n+1) - 6(n+1) + 2]}{2}$$

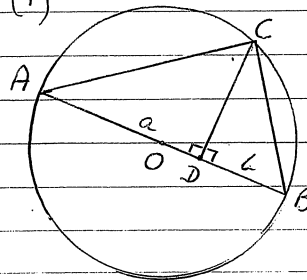
$$= \frac{n [6n^2 + 9n + 3 - 6n - 6 + 2]}{2}$$

$$= \frac{n (6n^2 + 3n - 1)}{2} \quad \left(\frac{1}{2}\right)$$

3

Question 8

(i)



$$\angle ACB = 90^\circ \text{ (angle in a semicircle)}$$

$$\text{Let } \angle CAD = \theta \therefore \angle BCD$$

$$\therefore \angle ACD = 90^\circ - \theta \text{ (angle sum of } \triangle ACD \text{ is } 180^\circ)$$

$$\angle CBA = 90^\circ - \theta \text{ (angle sum of } \triangle ABC \text{ is } 180^\circ)$$

$$\angle BCD = \theta \text{ (complement of } \angle ACD)$$

$$\triangle ACD \sim \triangle CBD \text{ (equiangular)}$$

$$\frac{CD}{BD} = \frac{AD}{CD} \text{ (corresponding sides in same ratio)}$$

$$CD^2 = AD \cdot BD$$

$$= a \cdot b$$

$$CD = \sqrt{ab} \quad (CD > 0)$$

$$(ii) \quad CD \leq \text{radius of circle} \quad \text{--- (i)}$$

$$\sqrt{ab} \leq \frac{a+b}{2}$$

$$(iii) \quad \therefore a+b \geq 2\sqrt{ab} \quad \text{for positive real numbers}$$

$$\therefore \text{If } x, y, z \text{ are positive real numbers}$$

$$\left. \begin{aligned} x+y &\geq 2\sqrt{xy} \\ y+z &\geq 2\sqrt{yz} \\ z+x &\geq 2\sqrt{zx} \end{aligned} \right\} \text{--- (i)}$$

$$\therefore (x+y)(y+z)(z+x) \geq 8\sqrt{xy \cdot yz \cdot zx}$$

$$= 8\sqrt{x^2 y^2 z^2}$$

$$= 8xyz \quad \text{--- (i)}$$

$$(u) \quad T_n = x^{n-1} (1 + x + x^2 + \dots + x^{n-1})$$

$$(i) \quad T_n = x^{n-1} \cdot \frac{(1-x^n)}{1-x}$$

$$= \frac{x^{n-1} - x^{2n-1}}{1-x} \quad \text{--- (1)}$$

$$S_n = T_1 + T_2 + T_3 + \dots + T_n$$

$$= \frac{1}{1-x} \left[1 + x + x^2 + \dots + x^{n-1} - (x + x^3 + x^5 + \dots + x^{2n-1}) \right]$$

$$= \frac{1}{1-x} \left[\frac{1(1-x^n)}{1-x} - \frac{x(1-x^{2n})}{1-x^2} \right] \quad \text{--- (1) } \begin{matrix} x \neq 1 \\ x^2 \neq 1 \end{matrix}$$

$$= \frac{1}{(1-x)} \frac{(1-x^n)(1+x) - x(1-x^{2n})}{(1-x^2)} \quad (1-x^n)(1+x^n)$$

$$= \frac{(1-x^n) [1+x-x(1+x^n)]}{(1-x)(1-x^2)}$$

$$= \frac{(1-x^n)(1-x^{n+1})}{(1-x)(1-x^2)} \quad \text{--- (1) } (x^2 \neq 1)$$

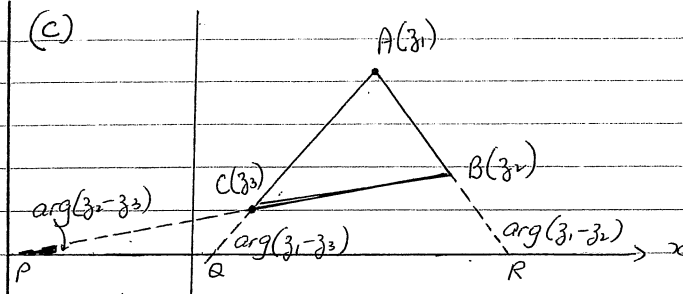
$$(ii) \quad \lim_{x \rightarrow 1} \frac{(1-x^n)(1-x^{n+1})}{(1-x)(1-x^2)} = \lim_{x \rightarrow 1} (T_1 + T_2 + T_3 + \dots + T_n)$$

$$= 1 + 2 + 3 + \dots + n \quad \text{--- (1)}$$

$$= \frac{n}{2} (1+n)$$

$$= \frac{1}{2} n(n+1) \quad \text{--- (1)}$$

(c)



① for diagram

$$\frac{z_2 - z_3}{z_1 - z_3} = \frac{z_1 - z_3}{z_1 - z_2} \quad \text{--- (1)}$$

Let AC meet x axis at Q	$\widehat{CQR} = \arg(z_1 - z_3)$
BC " " " P	$\widehat{CPQ} = \arg(z_2 - z_3)$
AB " " " R	$\widehat{BRx} = \arg(z_1 - z_2)$

$$\text{From (1) } \arg(z_2 - z_3) - \arg(z_1 - z_3) = \arg(z_1 - z_3) - \arg(z_1 - z_2)$$

$$\arg(z_1 - z_2) - \arg(z_1 - z_3) = \arg(z_1 - z_3) - \arg(z_2 - z_3)$$

$$\text{LHS} = \widehat{CAB} = \widehat{PCQ} = \text{RHS} \quad \text{--- (1)}$$

(exterior $\angle$ of ΔAQR) = sum of interior opp angles

(exterior $\angle$ of ΔPCQ) = sum of interior opp $\angle$ s

$$\widehat{PCQ} = \widehat{ACB} \quad (\text{vertically opp } \angle\text{s})$$

$$\therefore \widehat{ACB} = \widehat{CAB} (= \widehat{PCQ}) \quad \text{--- (1)}$$

Hence $AB = BC$ (equal sides opposite equal angles in a Δ)

$$|z_1 - z_2| = |z_2 - z_3| \quad \text{--- (2)} \quad \text{--- (1)}$$

$$\text{From (1) } \frac{|z_2 - z_3|}{|z_1 - z_3|} = \frac{|z_1 - z_3|}{|z_1 - z_2|}$$

$$|z_1 - z_3|^2 = |z_2 - z_3| |z_1 - z_2|$$

$$= |z_1 - z_2| |z_1 - z_2| \text{ from (2)}$$

$$\therefore |z_1 - z_3| = |z_1 - z_2| \quad \text{--- (1)}$$

$$\text{Hence } |z_1 - z_3| = |z_1 - z_2| = |z_2 - z_3| \text{ from (2)}$$

$$\therefore AC = AB = BC$$

ie $\triangle ABC$ is equilateral

There are other methods - each scored part marks for relevant facts that were established